

Section 2.4 Schöberl's thesis

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1 Overview of section 2.4

In section 2.4 of his thesis, Schöberl proceeds to apply the theory developed in section 2.3 to three particular sets of equations. The application of the theory of section 2.3 consists in:

1. Construct the primal system

$$A^\varepsilon(u, v) = f(v)$$

and the dual system

$$B^\varepsilon((u, p), (v, q)) = f(v).$$

2. Consider well-posedness of the system at $\varepsilon = 0$ in $V \times Q_0$ (Theorem 2.8). Remember that Q_0 is $\overline{\Lambda V}$ endowed with the norm:

$$\|p\|_{Q_0} \simeq \sup_{v \in V} \frac{c(p, \Lambda v)}{\|v\|}.$$

3. Define a norm $V \times Q$ and consider uniform well-posedness for $0 < \varepsilon \leq 1$ in $V \times Q$ (Theorem 2.9). The norm on Q is given by

$$\|p\|_Q^2 = \|p\|_{Q_0}^2 + \varepsilon \|p\|_c^2.$$

4. Choose elements for a non-conforming discretisation $V_h \times Q_h$ with bilinear form A_h^ε and prove stability of the discrete system.

2 Nearly incompressible materials

We analyse the equations from linear elasticity and consider the stability of the formulation in the incompressible limit. Note that the same primal formulation arises with Stokes when enforcing incompressibility with a penalty term [3] or with an augmented Lagrangian term [2].

2.1 Primal and dual formulations

The **primal** system for linear elasticity is set in

$$V = [H_{0,D}^1]^2$$

with the bilinear form

$$A^\varepsilon(u, v) = \underbrace{(e(u), e(v))_0}_{=:a(u,v)} + \varepsilon^{-1} \underbrace{(\operatorname{div} u, \operatorname{div} v)_0}_{=:c(\Lambda u, \Lambda v)}.$$

By defining $p = \varepsilon^{-1} \operatorname{div} u$, we get the **dual** formulation, defined in terms of the bilinear form

$$B^\varepsilon((u, p), (v, q)) = (e(u), e(v))_0 + (\operatorname{div} u, q)_0 + (\operatorname{div} v, p)_0 - \varepsilon(p, q)_0.$$

Thanks to Korn's inequality, which states that

$$\|u\|_1 \lesssim \|e(u)\|_0 \quad \forall u \in V$$

we have the coercivity of $a(\cdot, \cdot)$.

2.2 The Q_0 and Q norms

2.2.1 Pure Dirichlet boundary conditions

If we have pure Dirichlet boundary conditions, i.e. $\Gamma_D = \partial\Omega$, then

$$\|p\|_{Q_0} := \|p\|_{L^2/\mathbb{R}}$$

because

$$\|p\|_{L^2/\mathbb{R}} \lesssim \sup_v \frac{(p, \operatorname{div} v)}{\|v\|_1} \leq \|p\|_{L^2/\mathbb{R}}.$$

The lower bound follows from the inf-sup conditions on $L_0^2(\Omega) = L^2/\mathbb{R}$ and $V = H_0^1(\Omega)$. For the upper bound: given a constant $\alpha \in \mathbb{R}$,

$$\begin{aligned} \frac{(p, \operatorname{div} v)}{\|v\|_1} &= \frac{(p + \alpha, \operatorname{div} v)}{\|v\|_1} \leq \|p + \alpha\|_0 \\ \implies \sup_v \frac{(p, \operatorname{div} v)}{\|v\|_1} &\leq \inf_\alpha \|p + \alpha\|_0 = \|p\|_{L^2/\mathbb{R}}. \end{aligned}$$

Therefore, we have uniform well-posedness of B^ε in the norm $X = Q \times V$, with $Q = L_0^2(\Omega)$ with norm

$$\|p\|_Q^2 = \|p\|_{L^2/\mathbb{R}}^2 + \varepsilon \|p\|_0^2.$$

2.2.2 Mixed boundary conditions

If we have mixed boundary conditions (Dirichlet + Neumann), the inf-sup conditions hold over $Q = L^2(\Omega)$ (check) and we have

$$\|p\|_0 \lesssim \sup_v \frac{(p, \operatorname{div} v)}{\|v\|_1} \leq \|p\|_0$$

so $\|p\|_0 = \|p\|_{Q,0}$. This also implies that

$$\|p\|_0^2 \leq \|p\|_{Q,0}^2 + \varepsilon \|p\|_0^2 \leq 2\|p\|_0^2 \quad \forall \varepsilon \in (0, 1]$$

and therefore we have uniform well-posedness of B^ε on $X = V \times Q$, with $Q = L^2(\Omega)$ with its usual norm.

2.3 The discrete system

The $\mathbb{P}_2 - \mathbb{P}_0$ elements are used for the discrete velocity and pressure spaces $X_h = V_h \times Q_h$. Then, the discrete bilinear form $A_h^\varepsilon : V_h \times V_h \rightarrow \mathbb{R}$ is given by

$$A_h^\varepsilon(u_h, v_h) = (e(u_h), e(v_h))_0 + \varepsilon^{-1}(\overline{\operatorname{div} u_h}^h, \overline{\operatorname{div} v_h}^h)_0.$$

2.3.1 Stability in X_h

We can prove that the inf-sup conditions hold in X_h by constructing an appropriate Fortin operator. This is carried out in [1, Section 8.4.3].

Definition 1 (Fortin operator). *We call $I_h : V \rightarrow V_h$ a Fortin operator if*

1. *It is uniformly bounded, i.e.*

$$\|I_h v\| \lesssim \|v\| \quad \forall v \in V.$$

2. *It satisfies*

$$c(\Lambda I_h v, q_h) = c(\Lambda v, q_h) \quad \forall v \in V, \forall q_h \in Q_h.$$

For our case, property 2 can be rewritten as follows:

$$\begin{aligned} & (\operatorname{div}(I_h v), q_h)_0 = (\operatorname{div} v, q_h)_0 & \forall v, q_h \\ \iff & \sum_{K \in \mathcal{T}_h} \int_K \operatorname{div}(I_h v) q_h \, dx = \sum_{K \in \mathcal{T}_h} \int_K \operatorname{div}(v) q_h \, dx & \forall v, q_h \\ \iff & \int_K \operatorname{div}(I_h v - v) \, dx = 0 & \forall v, K \\ \iff & \int_{\partial K} (I_h v - v) \cdot n \, ds = 0 & \forall v, K. \end{aligned}$$

A standard way to construct such an operator (see [1, Proposition 5.4.4]) is by means of two operators I_h^1, I_h^2 from V onto V_h as follows:

Lemma 2. Let I_h^1, I_h^2 be two operators from V onto V_h with the following properties:

1. I_h^1 is uniformly bounded,

2. I_h^2 satisfies

$$c(\Lambda I_h^2 v, q_h) = c(\Lambda v, q_h)$$

(although in general it won't be uniformly bounded),

3. $I_h^2(I - I_h^1)$ is uniformly bounded.

Then

$$I_h^F = I_h^1 + I_h^2(I - I_h^1)$$

is a Fortin operator.

For the $\mathbb{P}_2 - \mathbb{P}_0$ elements, we set $I_h^1 : V \rightarrow V_h$ to be the standard Clement interpolator, which has the optimal approximation property

$$\sum_K h_K^{2r-2} |v - I_h^1 v|_{r,K}^2 \lesssim \|v\|_1^2 \quad r = 0, 1.$$

As a result, setting $r = 1$, using the triangle and Poincaré inequality, we see that I_h^1 is uniformly bounded:

$$\|I_h^1 v\|_1 \leq \|I_h^1 v - v\|_1 + \|v\|_1 \lesssim \|v\|_1.$$

We define $I_h^2 : V \rightarrow V_h$ by enforcing 6 (linearly independent) conditions for each component of the vector $I_h^2 v$ per triangle:

$$\begin{aligned} I_h^2 v|_K(M) &= 0 & \forall M \text{ vertex of } K, \\ \int_e I_h^2 v \, ds &= \int_e v \, ds & \forall e \text{ edge of } K. \end{aligned}$$

This last condition implies that

$$\int_{\partial\Omega} (I_h^2 v - v) \cdot n \, ds = 0$$

and therefore condition 2 holds for I_h^2 . We now need to check condition 3. We use a scaling argument to show that

$$|I_h^2 v|_{1,K} = |\widehat{I_h^2 v}|_{1,\hat{K}} \leq c \|\hat{v}\|_{1,\hat{K}} \leq c (h_K^{-1} \|v\|_{0,K} + |v|_{1,K}),$$

for $v \in V$ and $\hat{K} = h_K^{-1} K$ (so it has diameter 1 and the constants that arise in inequalities do not depend on h , only on minimum angles etc.). Now,

$$\|I_h^2(I - I_h^1)u\|_{1,K}^2 \leq c (h_K^{-2} \|(I - I_h^1)u\|_{0,K} + \|(I - I_h^1)u\|_{1,K}^2) \leq \|u\|_{1,K}^2.$$

3 Reissner-Mindlin plate

The first part of section 2.4.3 introduces the (primal) **scaled problem** and its dual counterpart. It also presents a regularity result, which motivates the introduction of the spaces V^+ , V^- and Q^+ and the characterisation of the dual of V^- .

3.1 The discrete stabilised bilinear form

We work directly with the discrete problem because a stabilised h dependent bilinear form is considered. A **stabilisation parameter** $\mu \in (0, 1]$ is introduced. The **material parameter** is

$$\varepsilon = \frac{t^2}{k}.$$

3.1.1 Primal formulation

We have the primal space

$$V = H_0^1(\Omega) \times [H_0^1(\Omega)]^2$$

which represents the space of transverse displacements (scalar) and rotations (vector). The space V_h is $\mathbb{P}_2 \times [\mathbb{P}_1^+]^2$. **Stabilised primal formulation:**

$$A_h^\varepsilon((w_h, \beta_h), (v_h, \delta_h)) = a((w_h, \beta_h), (v_h, \delta_h)) + \varepsilon^{-1} c(\Lambda_h(w_h, \beta_h), \Lambda_h(v_h, \delta_h)),$$

where

$$\begin{aligned} a((w_h, \beta_h), (v_h, \delta_h)) &= a^b(\beta_h, \delta_h) + \frac{k\mu}{h^2 + t^2} (\nabla w_h - \beta_h, \nabla v_h - \delta_h)_0, \\ \Lambda_h(w_h, \beta_h) &= \overline{\nabla w_h - \beta_h}^{-h}, \\ c(p_h, q_h) &= \left(1 - \frac{t^2\mu}{h^2 + t^2}\right) \|p\|_0^2. \end{aligned}$$

Here, $a^b(\cdot, \cdot)$ is uniformly continuous and coercive. Note that the bilinear forms depend on h and t . In order to have uniform continuity and coercivity for $a(\cdot, \cdot)$, we define the norm

$$\|(w, \beta)\|_V^2 := \|\beta\|_1^2 + \frac{1}{h^2 + t^2} \|\nabla w - \beta\|_0^2.$$

Regarding $\|\cdot\|_c$, uniform norm equivalence with $\|\cdot\|_0$ depends on the value of $\mu \in (0, 1]$:

$$\begin{aligned} \text{if } \mu < 1: \quad & (1 - \mu)\|p\|_0^2 \leq \|p\|_c^2 \leq \|p\|_0^2 \quad \implies \|p\|_0 \simeq \|p\|_c, \\ \text{if } \mu = 1: \quad & \|p\|_c^2 \simeq \min(1, h^2/t^2)\|p\|_0^2. \end{aligned}$$

For the last equivalence, note that if $\mu = 1$ then we have

$$1 - \frac{t^2}{t^2 + h^2} = 1 - \frac{1}{h^2/t^2 + 1} = \frac{h^2/t^2}{1 + h^2/t^2}.$$

Now, we consider the function $x/(1+x)$ for $x \geq 0$. We have

$$\frac{x}{1+x} \leq x \quad \text{and} \quad \frac{x}{1+x} \leq 1 \implies \frac{x}{1+x} \leq \min(1, x).$$

Also,

$$\begin{aligned} \text{if } 0 \leq x \leq 1: \quad & x(1+x) \leq 2x \implies \frac{1}{2}x \leq \frac{x}{1+x}, \\ \text{if } 1 \leq x: \quad & \frac{x}{1+x} \geq \frac{1}{2}, \end{aligned}$$

so

$$\frac{x}{1+x} \geq \frac{1}{2} \min(1, x).$$

3.1.2 Dual formulation

We introduce the dual variable

$$p = \varepsilon^{-1} \Lambda(w, \beta) = \frac{k}{t^2} (\nabla w - \beta)$$

which represents the scaled shear force. We have

$$Q = L^2(\Omega) \quad \text{and} \quad Q_h = \mathbb{P}_0.$$

We define the norm on Q such that we directly obtain stability:

$$\|p\|_Q^2 := \sup_{(w, \beta) \in V} \frac{c(\nabla w - \beta, p)^2}{\|(w, \beta)\|_V^2} + \varepsilon \|p\|_c^2.$$

References

- [1] D. Boffi, F. Brezzi, and M. Fortin, *Mixed finite element methods and applications*, Springer Series in Computational Mathematics, Springer, 2013.
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- [3] Qingguo Hong, Johannes Kraus, Jinchao Xu, and Ludmil Zikatanov, *A robust multigrid method for discontinuous Galerkin discretizations of Stokes and linear elasticity equations*, Numerische Mathematik **132** (2015).